MIL-M-13786(SigC)
Amendment No. 1
17 November 1955

MILITARY SPECIFICATION

MOTORS, FRACTIONAL HORSEPOWER, DIRECT CURRENT AND UNIVERSAL (FOR COMMUNICATION AND OTHER ELECTRONIC AND SPECIAL MILITARY APPLICATIONS)

1. Paragraph 2.1 Specifications

Delete:

72-53	Finishes (for Ground Signal Equipment).

Add:

MIL-B-3473	Brushes, Electrical Contact and Carbon plates, Electrical Contact Brush.
MIL-L-3545	Lubricating grease; high temperature.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-L-15719	Lubricating grease (High Temperature, Electric-motor, Ball and Roller Bearings).

STANDARDS

Add:

MS-91000	Brushes, Electrical Contact, Cartridge Type.

2. Paragraph 3.2.4 Lubricants. Delete entire paragraph and substitute:

"Lubricants shall conform to Specification MIL-G-3278, MIL-L-3545, MIL-L-15719 or MIL-L-6085."

3. Paragraph 3.3.1.1 Bearing mounting. Change to read as follows:

"Unless otherwise specified, when bearings are mounted in soft metal such as zinc alloys, aluminum or copper, a steel insert shall be provided."

4. Add the following paragraph:

"3.3.1.2 Heat stabilization. Bearings to be operated above 90° C shall be heat stabilized so that their dimensions do not increase more than 0.0001 inch per inch in diameter during 2500 hours at 149° C."

5. Add the following paragraph:

"3.3.1.3 Sleeve Bearings. Sleeve bearings shall have an oil reservoir adequate to permit 1000 hours of operation between lubrication periods."

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6. Paragraph 3.3.2 Impregnation. Delete the third sentence and add the following two sentences:

"For Class A and B insulation, impregnating varnish shall conform to Specification MIL-V-1137, Grade CB, unless otherwise specified by the contracting officer. For Class H insulation, silicone impregnating varnish shall be used."

7. Add the following paragraph:

"3.3.3. Armature. - Motors shall be so constructed that the armature can be readily removed without the use of special tools."

8. Add the following paragraph:

"3.9.2.3 Tapped holes. Where tapped holes are used for mounting or assembly, suitable protection shall be included in the design of the motor to prevent damage to the motor winding should an extra long screw be used."

9. Paragraph 3.9.4.1 Brush material and design. In first sentence after "Brushes shall be" add "in accordance with Specification MIL-B-3743 or Military Standard MS91000, and shall be"

10. Paragraph 3.9.4.2 Spare brushes. Delete first sentence and substitute, "Unless otherwise specified on the complimentary document, one set of spare brushes shall be supplied with each motor."

11. Paragraph 3.9.4.3 Brush holders. Add as second sentence, "The brush holders shall be of such design that assembly and disassembly of the motor can be accomplished without loss of brush setting indication."

12. Add the following paragraph:

"3.9.4.5 Brush life. Brushes used in motors designed for sea level operation shall have a useful life of 1000 hours. Brushes used in motors designed for air-borne operation shall have a useful life of at least 1,000 hours when operated at altitudes varying from sea level to 35,000 feet. When operating at altitudes varying from 35,000 feet to 50,000 feet, the useful life of the brushes shall be at least 500 hours."

13. Add the following paragraph:

"3.9.5.1 Commutator step. A turned down section (Step to show allowable wear and minimum resurfacing diameter), 0.030 inch wide minimum by a depth of 1.5 percent/minimum of the commutator diameter, but not less than 0.015 inch, shall be provided at the outboard end of the commutator. A milled slot in the end of the bars is permissible, in place of the step, on commutators 2 inches in diameter or under."

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14. Paragraph 3.9.9.1 Continuous. Add as last sentence, "Unless otherwise specified, the motor shall meet the requirements for full-load speed and torque after the 2500 hour life test specified in paragraph 3.2.21. Commutator wear shall not exceed 50% of the allowable wear (see 3.9.5.1)."

15. Paragraph 3.9.9.2 Intermittent. Add as last sentence, "Unless otherwise specified, the motor shall meet the requirements for full-load speed and torque after the 1000 hour life test specified in paragraph 4.2.21.2. Commutator wear shall not exceed 75% of allowable wear. (See 3.9.5.1)."

16. Paragraph 3.23 Moisture Resistance. Delete "48 hour drying period", and substitute, "24 hour running period".

17. Paragraph 3.26 Marking. Delete paragraph and substitute:

"The motor shall be permanently and legibly marked with the designation "MOTOR, DIRECT CURRENT" or "MOTOR, UNIVERSAL" as applicable, the government drawing of specification sheet number, the manufacturer's name or symbol, horsepower, duty, cycle, full-load speed, voltage, full-load current, direction or rotation, serial no., and wiring diagram. Terminal marking shall conform to Specification JAN-E-251. The direction of rotation shall be plainly and permanently indicated by an arrow for unidirectional and reversible motors."

18. Paragraph 3.28 Delete Item 13 "Life".

19. Add the following:

3.30 Altitude. When tested in accordance with 4.2.18, the cumulative brush or commutator wear shall not exceed 20 percent of the allowable wear (see 3.9.5.1) during steps 1 and 2 of this test. Wear during step 3 shall not exceed 4 percent of the allowable wear. The temperature rise values at the end of step 2 and 3 shall not exceed the seal level value (step 1) by more than 50%."

20. Paragraph 4.1.2 Standard Test conditions. After "herein" add, "or on the complimentary document."

21. Paragraph 4.2.3.2 Full load. After "Full load". Delete the sentence following.

22. Paragraph 4.2.3.5 No load. Delete first sentence.

23. Paragraph 4.2.3.6 Effect of variation in line voltage. Delete first sentence.

24. Paragraph 4.2.5 Overspeed. Delete first sentence.

25. Paragraph 4.2.6 Insulation resistance. Change paragraph to read as follows:

"Measurements shall be made at 500 volts dc. The test voltage shall be applied between ground (frame, positive instrument lead) and each winding

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(negative instrument lead) which is not permanently grounded by soldering, brazing, or welding, with all other windings grounded. Windings which are normally connected together in service shall be considered as one winding for the purpose of this test. A second reading using reversed instrument polarity (negative instrument lead grounded and positive instrument lead to the winding) shall be made. The lowest reading shall be considered the insulation resistance. (See 3.14).

26. Add the following paragraph:

"4.2.8.1 Air gap measurements. For motors having a rating above 1/4 horsepower, the uniformity of the air-gap shall be determined by ascertaining that the rotor turns freely in the assembled motor, when wound with a wire spaced spirally around the rotor periphery. For this test, the diameter of the wire shall be 70 percent of the nominal air-gap of the motor."

27. Paragraph 4.2.11 Temperature rise. After "Temperature rise." Delete the sentence following.

28. Paragraph 4.2.13 Vibration. Under "(c) Motion" Change: "7-1/2 hours" to "6 hours", "2-1/2 hours" to "2 hours", and "1-1/4 hours" to "1 hour".

29 Paragraph 4.2.15 Overload. Delete first sentence.

30. Paragraph 4.2.17.1.2 Number of blows. Last sentence, change height of hammer blows from: "2 feet, 3 feet and 4 feet" to "1 foot, 3 feet and 5 feet".

31. Paragraph 4.2.18 High altitude operation. (When applicable). Delete entire paragraph and substitute:

"The motor shall be operated under the following conditions. New brushes may be installed for this test. The motor shall be operated prior to each altitude run for a minimum of 1 hour but not more than 2 hours at rated load and duty cycle, under standard test conditions. (See 4.1.2). The temperature rise of the motor winding shall be determined at the end of the last cycle of each step. (See 3.14 and 4.2.6). The rate of change in altitude need not be controlled. The maximum dew point for all tests shall be at -60° C.

Step 1. Operate the motor for 100 hours at rated load and duty cycle at standard test conditions.

Step 2. Operate the motor at rated load and duty cycle for four continuous cycles consisting of the following: Two cycles of 9 hours each at altitude conditions approximately 35,000 feet and -55° ± 5° C (-67° ± 9° F) followed by two cycles of 9 hours each at altitude conditions approximately 35,000 feet and -10° ± 5° C (+14 ±9° F).

Step 3. Operate the motor at rated load and duty cycle for two continuous cycles consisting of the following: One cycle of 9 hours at altitude conditions approximately 50,000 feet and -55° ±5° C (-67° ±9° F) followed by one cycle of 9

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hours at altitude conditions approximately 50,000 feet and -10° ± 5°C (14° ± 9°F).

32. Paragraph 4.2.19(d) Moisture resistance. Fourth sentence change: "One hour at full-load torque, "to "24 hours at full-load torque," after "full-load speed," add "and insulation resistance". Delete 5th sentence.

33. Paragraph 4.2.21, 4.2.21.1 and 4.2.21.2. Delete and substitute:

"4.2.21 Life. During life test, precautions shall be taken against undue air contamination of a type not to be expected in normal service use. The motors shall be operated at rated voltage and output torque at their specified duty cycle. Continuous duty motors shall be operated for 2500 hours and intermittent duty motors shall be operated for 1000 hours. Unless otherwise specified on the complimentary document, the life test shall be conducted at an ambient temperature of 65°C (150°F). Unidirectional continuous duty motors shall be stopped and started every 24 hours. Reversible continuous duty motors shall be reversed every 24 hours. Reversible intermittent duty motors shall be alternately operated in each direction at rated duty cycle. Following this test, the motor shall pass the full load speed and torque test at ambient room temperature. No maintenance of any kind, including dressing of the commutator shall be allowed during this test except that brushes may be changed at not less than 1000 hour intervals, and sleeve bearings may be lubricated at not less than 1000 hour intervals."

34. Paragraph 4.2.22(b) Corrosion. 2nd line, after, "Subjected to", add: "the full load test at".

35. Paragraph 6.7 Frame sizes. Change title of paragraph and first sentence to read as follows:

"6.7 NEMA frame sizes. In order to accomplish standardization of motors for military applications, motors should conform to the NEMA Standard frame sizes indicated in figure 3 wherever possible."

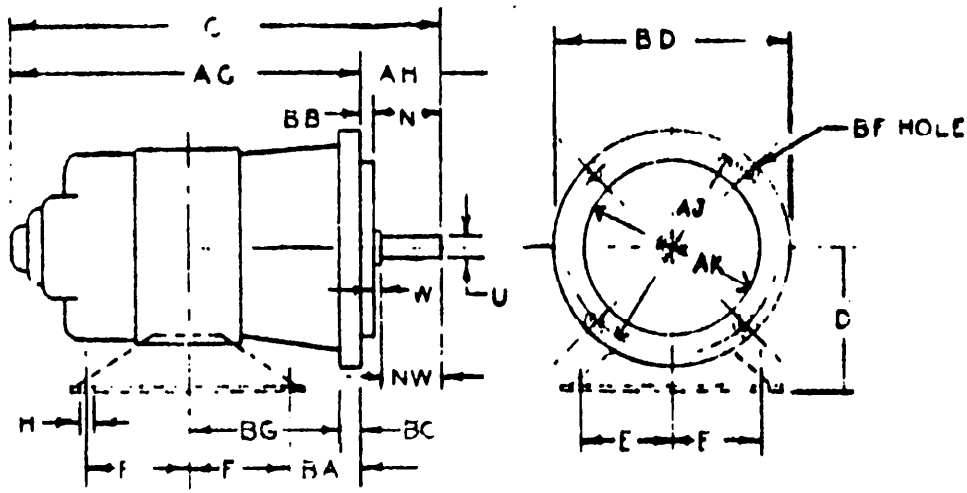
36. Figure 2 Motor Identification Form. Add as item 30:

"30, Specialized test conditions (if applicable)"

37 Figure 3 Change item under title from "(From NEMA Standard MG-1-3.2)" to "(From NEMA Standard MG-1-3.02)"

38. Add Figure 3a.

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Frame No.	U	AH	AJ	AK	BB	BC	BC Max.	BG
42C	3/8	1-5/16	3-3/4	3	5/32Max	3/16	5	2-23/32
48C	1/2	1-11/16	3-2/4	3	5/32 "	3/16	5-5/8	3-11/16
56C	5/8	2-1/16	5-7/8	4-1/2	5/32 "	3/16	6-1/2	4-1/16
66C	3/4	2-7/16	5-7/8	4-1/2	5/32 "	3/16	6-1/2	5-7/16

BF Hole

Frame No.	Number	Tap Size
42C	4	1/4-20
48C	4	1/4-20
56C	4	3/8-16
66C	4	3/8-16

All Dimensions in inches
TOLERANCES
AK Dimension- 42-66 frames, incl. - +0.000, -0.003 inch.
Face Runout - 42-66 frames, incl. - -0.004 inch indicator reading.
Permissible Eccentricity of Mounting Rabbet - 42-66 frames, incl. - -0.004 inch indicator reading.

Notes:
W-W is the length of shaft extension from shoulder to shaft end.

Standard Dimensions for Type C Face-mounted, Direct and Alternating-current Motors
(From NEMA Standard MG 1-3.06)
FIG 3a

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